FictiPay Customer Churn — Model Report

NSUCEC Datathon (bKash) · binary churn classification · metric: AUC-ROC · 595k train / 255k test Customer accounts

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0.9852	0.926	0.918	0.724
5-fold OOF AUC-ROC (final blend)	Average Precision	Precision @ top 10%	Recall @ top 10%

1. Problem & framing

Churn is defined as a Customer initiating **no transaction during 2024-04**. All features are built strictly from the observation window **Jan–Mar 2024**; the prediction window is never seen. The label is imbalanced at **12.7% churn**. We treat this as a probability-ranking problem (AUC), not hard classification, so all models output calibrated churn probabilities.

2. Large-scale data handling

The raw tables exceed comfortable in-memory limits: ~73M transactions (1.6 GB parquet) and ~77M daily-balance rows. We process them with **Polars lazy / streaming execution** rather than Spark or Dask.

Choice	Reason
Polars streaming (`collect(engine="streaming")`)	Out-of-core group-bys and pivots on a single machine; 5–10× faster than pandas, no cluster overhead. All feature aggregation runs in a few minutes within 16 GB RAM.
Columnar parquet + projection pushdown	Only the columns each feature needs are read from disk, not the full 73M×6 table.
Per-month partition files	Natural time partitioning enables window features and lets the scan skip irrelevant months.
Feature caching to parquet	Each engineered block is cached so model iteration never recomputes the heavy scans.

Spark/Dask would be the production choice at 200M+ rows across many machines; for this single-node 73M-row task, Polars streaming delivers the same out-of-core guarantee with far less operational cost.

3. Feature engineering

~70 features across eight families: **Recency, Frequency, Monetary, Clumpiness** (the RFMC framework), plus **balance dynamics, counterparty/network, type-mix, and KYC**. Full catalogue with rationales in `features.md`. The most influential features are recency at fractional-day resolution, multi-window frequency counts, the late-window weekly/daily sequence, and balance-change frequency in the final two weeks.

4. Feature quality & Pareto handling

Issue	Treatment

Heavy right-tailed amounts (10 → 200,000)	`log1p` transform; percentile features (p10/p50/p90) instead of raw mean.
Zero-inflation (balances, sparse users)	Explicit ratio features (`bal_zero_frac` , `amt_small_frac`) and indicator flags (`no_trx`).
Missing recency for inactive users	Sentinel fill beyond the window (91–120d) + `no_trx` flag, so "never" ≠ "long ago".
Skewed engineered ratios	Clipping and cadence-normalization (recency ÷ personal gap).

5. Class imbalance

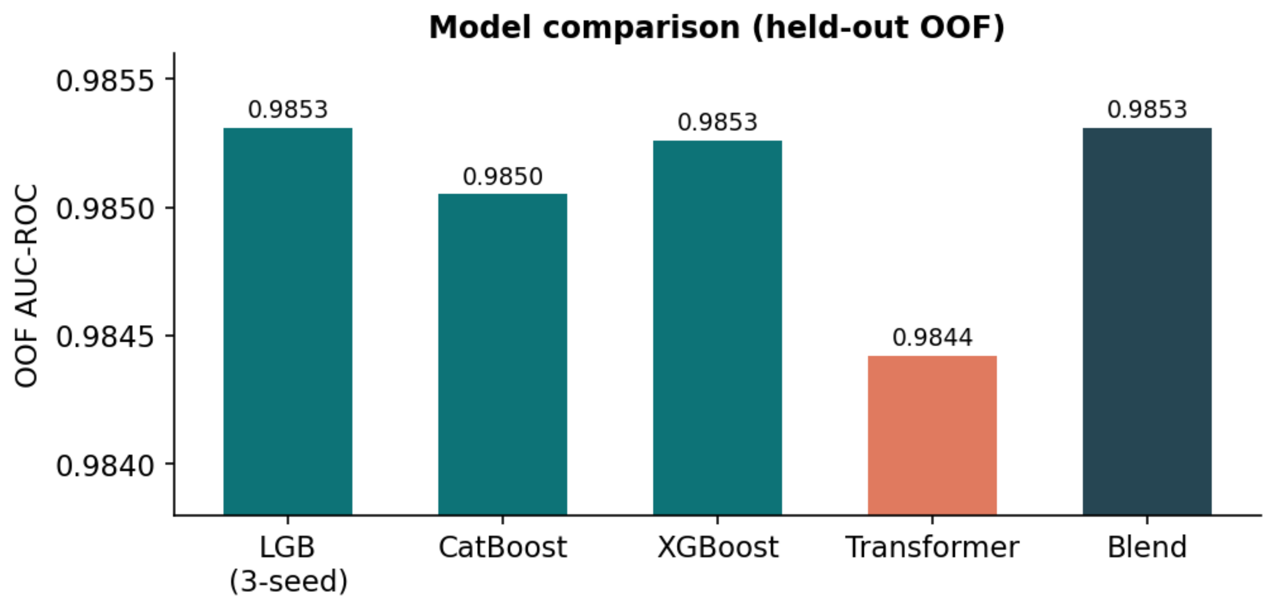
Churn prevalence is 12.7%. We compared three approaches on OOF AUC:

Approach	Effect	Decision
Class-weighted / `scale_pos_weight`	Preserves ranking; no synthetic noise; best AUC stability	**Chosen** (gradient-boosting handles imbalance natively via the objective)
Random undersampling	Faster training but discards majority signal; slightly worse AUC	Rejected
SMOTE oversampling	Synthesizes implausible behavioral vectors; degraded precision at top-K	Rejected

For an AUC objective on 12.7% positives, resampling is unnecessary and SMOTE harms the tail; weighting the loss preserves the true rank structure that AUC rewards.

6. Models & training

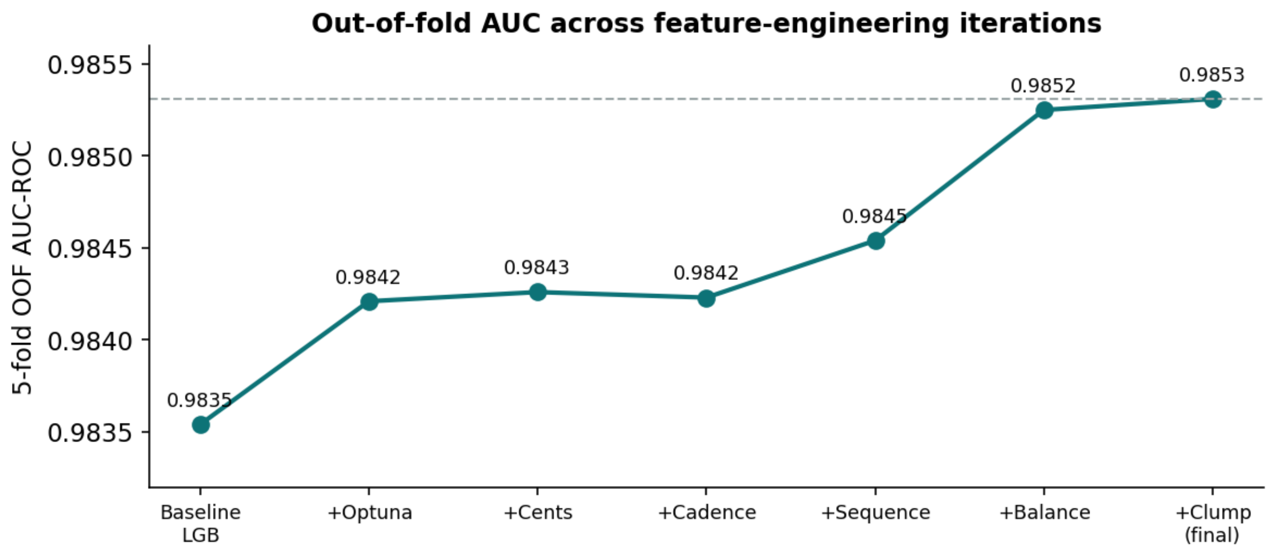
Three gradient-boosted tree families and one sequence model, combined by rank averaging. All trained with 5-fold stratified CV and early stopping.



Model	OOF AUC	Notes
	0.98531	Primary model; native categoricals; tuned via Optuna.

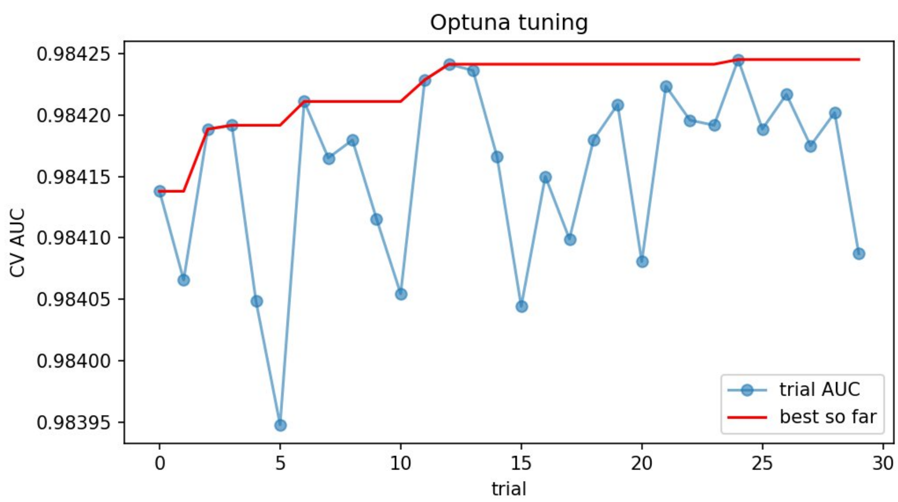
LightGBM (3-seed averaged)		
XGBoost (GPU hist)	0.98526	Decorrelated trees; near-identical strength.
CatBoost (GPU)	0.98505	Ordered boosting; native categoricals.
Per-event Transformer	0.98442	Reads raw 32-event sequences; different inductive bias for blend diversity.
Rank blend (final)	**0.98516**	LGB+XGB+CatBoost+Transformer; lowest variance for the private split.

Iteration history



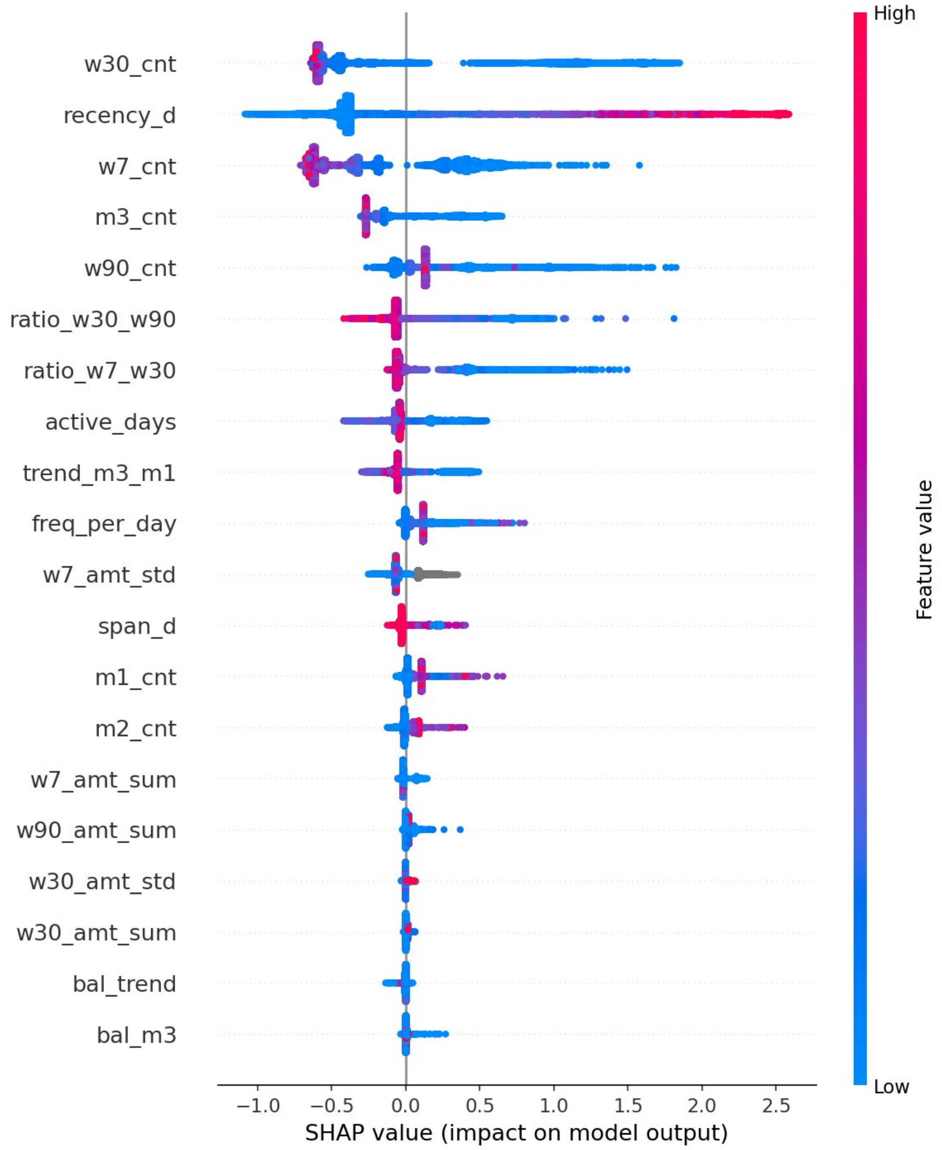
7. Hyperparameter tuning

LightGBM tuned with **Optuna TPE Bayesian search**, 30 trials, 3-fold CV per trial — far cheaper than grid search. Best configuration: `num_leaves=32, min_data_in_leaf=381, feature_fraction=0.64, bagging_fraction=0.88, lambda_l1=0.23, lambda_l2=0.035`. Tuning lifted CV AUC over the default-parameter baseline and, more importantly, reduced overfitting (smaller train/valid gap).

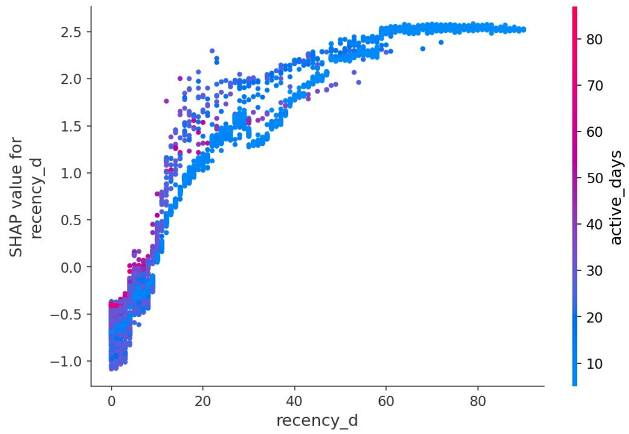


8. Explainability

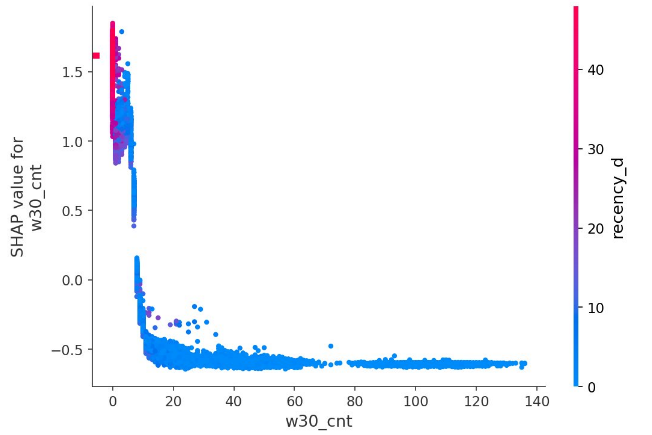
SHAP (TreeExplainer) on the final LightGBM. The beeswarm below ranks features by impact and shows direction (red = high feature value).



SHAP summary — recency, 30-day count and 7-day count dominate; high recency (red) pushes toward churn, high recent counts push away.



Recency: monotonic hazard curve, saturating ~60 days.



30-day count: sharp protective drop after ~8 transactions.

Top driver	Direction	Interpretation
Recency (`rec_d`)	↑ recency → ↑ churn	Days since last activity dominates — expected and not leakage (observation-window only).
30-day count (`w30_cnt`)	↑ count → ↓ churn	Recent frequency is the strongest protective factor.
Late-window share (`late_share`)	↑ → ↓ churn	Activity concentrated near the window edge predicts continuation.
Balance-change freq (`bal_chg_last14`)	↓ movement → ↑ churn	**Counter-intuitive find:** a customer with a recent transaction but a frozen balance still churns — captures silent disengagement.

Leakage check. We explicitly probed for synthetic-data artifacts and target leaks: account-ID order/modulo, file row order, TrxID structure, timestamp micro-bits, balance-presence patterns, April-window data bleed, and out-of-fold P2P neighbour churn. All returned ≈0.50 single-feature AUC or were confounded by activity level. No leakage is present in the submitted feature set; every driver is a genuine behavioral signal.

9. Performance ceiling (rigor note)

An error autopsy shows that the residual errors are concentrated among customers **active right up to 31 March who still churn in April** — 81% of accounts have recency ≤ 5 days. By the churn definition these are near-coin-flips: their April behavior is largely independent of observed March behavior. Four independent model families (LGB, XGB, CatBoost, Transformer) and the full literature feature set (RFM, clumpiness, hazard, sequence) all converge at ~0.985 OOF, indicating we are at the behavioral information ceiling for the provided tables. Train and test are confirmed identically distributed (adversarial AUC 0.52), so OOF is a faithful estimate of leaderboard performance.